INCH-POUND

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SUPERSEDING
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# DETAIL SPECIFICATION

# CLOTH, PLAIN WEAVE, RIPSTOP, NYLON

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

## 1. SCOPE

1.1 Scope. This specification covers nylon, plain weave, ripstop cloth.

1.2 Classification. This specification covers the following classes as specified (see 6.2).

1.2.1 Classes.

| | |
|---|---|
| Class 1 | Dyed |
| Class 2 | Natural |
| Class 3 | DELETED (see 6.6) |
| Class 4 | Woodland Camouflage Pattern |
| Class 5 | Desert Camouflage Pattern |
| Class 6 | DELETED (see 6.6) |
| Class 7 | Operational Camouflage Pattern (OCP) |
| Class 8 | Woodland MARPAT |

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Sections 3 and 4 of this specification whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

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2.2 Government documents.

2.1.1 Specifications, standards, and handbooks. Not Applicable

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND SOLDIER CENTER

| | |
|---|---|
| 2-1-1516 | - Woodland Camouflage Pattern |
| 2-1-2240 | - Desert Pattern |
| 2-1-2525 | - Woodland MARPAT – 4 color (Coyote 476) |
| 2-1-2526 | - Woodland MARPAT – 4 color (Green 474 with EGA symbol) |
| 2-1-2527 | - Woodland MARPAT – 4 color (Black 477) |
| 2-1-2528 | - Woodland MARPAT – 4 color (Khaki 476) |
| 2-1-2592 | - Operational Camouflage Pattern (OCP). |

(Comments, suggestions, or questions on this document should be addressed to: Department of the U.S. Army, DEVCOM - Soldier Center, ATTN: FCDD-SCD-PE, 10 General Greene Avenue, Natick, MA 01760-5019.)

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

| | |
|---|---|
| AATCC EP1 | - Evaluation Procedure for Gray Scale for Color Change |
| AATCC EP2 | - Evaluation Procedure for Gray Scale for Staining |
| AATCC EP8 | - Evaluation Procedure for AATCC 9 Step Chromatic Transference Scale Rating |
| AATCC EP9 | - Evaluation Procedure for Visual Assessment of Color Difference of Textiles |
| AATCC TM8 | - Test Method for Colorfastness to Crocking: AATCC Crockmeter Method |
| AATCC TM15 | - Test Method for Colorfastness to Perspiration |
| AATCC TM20 | - Test Method for Fiber Analysis: Qualitative |
| AATCC TM61 | - Test Method for Colorfastness to Laundering: Accelerated |



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AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles

(Copies of these documents are available online at https://www.aatcc.org.)

ASTM INTERNATIONAL

| | |
|---|---|
| ASTM D1776/D1776M | - Standard Practice for Conditioning and Testing Textiles |
| ASTM F2203 | - Standard Test Method for Linear Measurement Using Precision Steel Rule |
| ASTM D2262 | - Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Method (Constant-Rate-of- Traverse Tensile Testing Machine) |
| ASTM D3775 | - Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics |
| ASTM D3776/D3776M | - Standard Test Method for Mass per Unit Area (Weight) of Fabrics |
| ASTM D3990 | - Standard Terminology Relating to Fabric Defects |
| ASTM D5035 | - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method) |

(Copies of these documents are available online at https://www.astm.org.)

OTHER PUBLICATIONS

Fabric Defect Replica Scales

(For Information on obtaining copies of the Fabric Defect Replica Scales please contact U.S.Army, DEVCOM – Soldier Center, ATTN: FCDD-SCP-PE, 10 General Greene Avenue, Natick, MA 01760-5019.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## 3 REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Conformance inspection. When specified (see 6.2), a sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance, and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Figures. Figures 1 through 3 are furnished for informational purposes only. To the extent of any inconsistencies between the written document and the figure, the written document shall govern.

3.5 Materials.

3.5.1 Yarn. The yarn used shall be multifilament nylon.

3.6 Color.

3.6.1 Visual shade matching (all Classes). The color and appearance of the cloth shall match the standard sample when tested as specified in Table IX (see 4.4.6).

3.6.2 Colorfastness. The finished cloth shall conform to the colorfastness requirements listed below in Table I unless otherwise specified in the contract or procurement documents when tested as specified in 4.4.6 and 4.5.2.

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TABLE I. Colorfastness requirements (Classes 1, 4, 5, 7, and 8).

| Camouflage Pattern | Color Evaluation | Laundering Color Change & Staining (3 cycles) (min.) | Perspiration (Acid & Alkaline) Color Change & Staining (min.) | Crocking Dry/Wet (min.) |
|---|---|---|---|---|
| Dyed (Class 1) | All colors | 3-4 | 3-4 | 3.5 |
| Woodland Camouflage (Class 4) | All colors except black 357 | 3-4 | 3-4 | 3.5 |
| | Black 357 | 3-4 | 3-4 | 1.5 |
| Desert Camouflage (Class 5) | All colors | 3-4 | 3-4 | 3.5 |
| Operational Camouflage (Class 7) | All colors | 3-4 | 3-4 | 3.5 |
| Woodland MARPAT (Class 8) | All Colors, except for Black 477 | 3-4 | 3-4 | 3.5 |
| | Black 477 | 3-4 | 3-4 | 1.5 |

3.6.3 Dyeing/printing of cloth for Class 1, Dyed (solid shades). The Class 1 finished cloth shall be dyed in one (1) of the following solid shades: Camouflage Green 483, Foliage Green 504, Urban Gray 505, Coyote 498, Coyote Brown 3758, Tan 380, Tan 499, and Arctic White, or as otherwise specified in the contract or purchase order (see 6.2), and shall match the standard sample (see 6.4).

3.6.4 Dyeing/printing of cloth for Class 2. The cloth shall be natural (unbleached).

3.6.5 Dyeing/printing of cloth for Class 4. The cloth shall be dyed to a ground shade to match the Light Green 354 and then it shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Light Green 354, the remaining three (3) colors shall be printed as appropriate, for the Dark Green 355, Brown 356, and Black 357 areas of the pattern. When the ground shade is not dyed to approximate Light Green 354 all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.6.6 Dyeing/printing of cloth for Class 5. The cloth shall be dyed to a ground shade to match the Light Tan 492 and then it shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Light Tan 492, the remaining two (2) colors shall be printed as appropriate, for the Light Brown 493, and Light Khaki 494. When the ground shade is not dyed to approximate Light Tan 492 all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.6.7 DELETED



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3.6.8 Dyeing/printing of cloth for Class 7. The cloth shall be dyed to a ground shade matching or approximating Dark Cream 559, and then it shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Dark Cream 559, the remaining six (6) colors shall be printed as appropriate, for the Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529 and Bark Brown 561 areas of the pattern. When the ground shade is not dyed to approximate Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.6.9 Dyeing/printing of cloth for Class 8. The cloth shall be dyed to a ground shade matching or approximating Khaki 475 and then overprinted with the camouflage pattern. When the ground shade is dyed to match Khaki 475, the remaining three (3) colors shall be printed as appropriate, for the Green 474, Coyote 476 and Black 477 areas of the pattern. When the ground shade is dyed to approximate Khaki 475, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.7 Pattern execution, Class 4, 5, 7, and 8. The pattern on the printed finished cloth(s) shall match the standard sample with respect to design, colors, and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage. Skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When a standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.4), and the pattern on the finished cloth shall match that of the drawing (see 2.2.2. and 6.2). The pattern repeat for each Class shall be as follows:

Class 4 Woodland Camouflage Pattern - 27.25 (+1.25, -2.50) inches in the warp direction.
Class 5 Desert Camouflage Pattern - 16.75 (+1.25, -1.75) inches in the warp direction.
Class 7 Operational Camouflage Pattern - 25.25 (+1.25, -2.50) inches in the warp direction.
Class 8 Woodland MARPAT - 35.00 (±1.50) inches in the warp direction.

3.7.1 Pattern width (filling direction for OCP, Class 7). The maximum camouflage printed area for OCP, Class 7 shall be 68-inches in the filling direction.

3.8 Spectral reflectance. The spectral reflectance values for each Class shall conform to the requirements specified in their respective applicable Tables II-A through VI when tested as specified in 4.5.4.

3.8.1 Class 1, Solid shades. The reflectance values for Class 1 shall conform to the requirements listed below, in Table II-A, when tested as specified in 4.5.4.

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TABLE II-A. Spectral reflectance (percent), Class 1.

| Solid Shades | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Wave-Length (nm) | Camouflage Green 483 | | Foliage Green 504 | | Urban Gray 505 | | Coyote 498 | | Coyote Brown 3758 | | Tan 380 | | Tan 499 | |
| | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| 600 | 3 | 10 | 8 | 26 | 12 | 26 | 8 | 20 | 8 | 20 | N/A | N/A | 8 | 26 |
| 620 | 3 | 10 | 8 | 26 | 14 | 26 | 8 | 20 | 8 | 20 | N/A | N/A | 8 | 26 |
| 640 | 3 | 10 | 8 | 28 | 14 | 28 | 8 | 22 | 8 | 22 | N/A | N/A | 8 | 30 |
| 660 | 3 | 11 | 10 | 30 | 14 | 30 | 8 | 24 | 8 | 26 | N/A | N/A | 8 | 34 |
| 680 | 3 | 13 | 10 | 34 | 18 | 34 | 12 | 24 | 10 | 27 | N/A | N/A | 12 | 38 |
| 700 | 4 | 28 | 12 | 38 | 24 | 38 | 12 | 34 | 12 | 53 | 25 | 53 | 12 | 40 |
| 720 | 5 | 40 | 16 | 42 | 26 | 42 | 16 | 42 | 16 | 54 | 25 | 54 | 16 | 46 |
| 740 | 7 | 52 | 16 | 46 | 30 | 46 | 22 | 46 | 20 | 55 | 25 | 55 | 22 | 50 |
| 760 | 11 | 60 | 18 | 48 | 32 | 48 | 30 | 50 | 21 | 56 | 26 | 56 | 30 | 50 |
| 780 | 17 | 64 | 18 | 48 | 34 | 48 | 34 | 54 | 21 | 57 | 27 | 57 | 34 | 54 |
| 800 | 24 | 67 | 20 | 50 | 34 | 50 | 36 | 56 | 22 | 58 | 28 | 58 | 36 | 56 |
| 820 | 32 | 70 | 22 | 54 | 36 | 54 | 38 | 58 | 23 | 59 | 30 | 59 | 38 | 58 |
| 840 | 37 | 71 | 24 | 54 | 38 | 54 | 38 | 58 | 24 | 62 | 33 | 62 | 38 | 58 |
| 860 | 40 | 73 | 26 | 56 | 40 | 56 | 40 | 60 | 25 | 65 | 36 | 65 | 40 | 60 |

3.8.1.1 Class 4, Woodland Camouflage. The reflectance values shall conform to the requirements listed below, in Table II-B, when tested as specified in 4.5.4.

TABLE II-B. Spectral reflectance requirements, Class 4.

| Reflectance Values (percent) - Woodland Camouflage | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nm) | Light Green 354 | | Dark Green 355 & Brown 356 | | Black 357 | |
| | Min. | Max | Min. | Max | Min. | Max |
| 600 | 8 | 20 | 3 | 9 | N/A | N/A |
| 620 | 8 | 20 | 3 | 9 | N/A | N/A |
| 640 | 8 | 20 | 3 | 9 | N/A | N/A |
| 660 | 8 | 20 | 3 | 12 | N/A | N/A |
| 680 | 10 | 30 | 3 | 16 | N/A | N/A |
| 700 | 18 | 50 | 5 | 32 | N/A | 20 |
| 720 | 22 | 54 | 7 | 44 | N/A | 30 |
| 740 | 30 | 56 | 12 | 52 | N/A | 33 |
| 760 | 35 | 58 | 18 | 56 | N/A | 33 |
| 780 | 40 | 62 | 26 | 56 | N/A | 34 |
| 800 | 55 | 80 | 34 | 56 | N/A | 34 |
| 820 | 55 | 80 | 42 | 60 | N/A | 35 |
| 840 | 55 | 84 | 44 | 60 | N/A | 35 |
| 860 | 60 | 84 | 44 | 60 | N/A | 35 |



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3.8.2 Class 5, Desert Camouflage. The reflectance values shall conform to the requirements listed below, in Table III, when tested as specified in 4.5.4.

TABLE III. Spectral reflectance requirements, Class 5.

| Reflectance Values (percent) - Desert Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, nanometers (nm) | Light Tan 492 | | Light Brown 493 | | Light Khaki 494 | |
| | Min | Max | Min | Max | Min | Max |
| 700 | 38 | 53 | 19 | 36 | 25 | 48 |
| 720 | 38 | 58 | 20 | 36 | 25 | 52 |
| 740 | 39 | 62 | 20 | 36 | 25 | 54 |
| 760 | 40 | 66 | 21 | 36 | 26 | 56 |
| 780 | 41 | 72 | 21 | 38 | 27 | 57 |
| 800 | 43 | 76 | 22 | 43 | 28 | 58 |
| 820 | 45 | 76 | 23 | 45 | 30 | 58 |
| 840 | 48 | 78 | 24 | 46 | 33 | 58 |
| 860 | 50 | 78 | 25 | 46 | 36 | 59 |

3.8.3 and Table IV. DELETED

3.8.4 Class 7, Operational Camouflage. The reflectance values shall conform to the requirements listed in Table V, when tested as specified in 4.5.4.

TABLE V. Spectral reflectance requirements, Class 7.

| Reflectance values (percent) - Operational Camouflage Pattern (OCP) | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength (nm) | Dark Cream 559 & Tan 525 | | Light Sage 560, Olive 527 & Brown 529 | | Dark Green 528 & Bark Brown 561 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 13 | 25 | 7 | 18 | 3 | 11 |
| 620 | 13 | 25 | 7 | 18 | 3 | 12 |
| 640 | 14 | 26 | 7 | 18 | 3 | 12 |
| 660 | 15 | 27 | 8 | 20 | 4 | 13 |
| 680 | 20 | 30 | 9 | 22 | 5 | 14 |
| 700 | 29 | 39 | 10 | 23 | 5 | 14 |
| 720 | 36 | 47 | 12 | 24 | 5 | 14 |
| 740 | 43 | 54 | 17 | 30 | 8 | 17 |
| 760 | 51 | 62 | 29 | 40 | 11 | 22 |
| 780 | 55 | 65 | 33 | 44 | 14 | 24 |
| 800 | 56 | 67 | 34 | 45 | 15 | 25 |
| 820 | 56 | 67 | 35 | 46 | 15 | 25 |
| 840 | 57 | 68 | 35 | 46 | 15 | 25 |
| 860 | 57 | 68 | 35 | 46 | 15 | 25 |



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3.8.5 Class 8, Woodland MARPAT. The reflectance values shall conform to the requirements listed in Table VI, when tested as specified in 4.5.4.

TABLE VI. Spectral reflectance requirements, Class 8.

| Reflectance Values (percent) - Marine Pattern (MARPAT) Woodland Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelengths Nanometers (nm) | Coyote 476 & Khaki 475 | | Green 474 | | Black 477 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 10 | 20 | 3 | 9 | N/A | 10 |
| 620 | 10 | 20 | 3 | 9 | N/A | 10 |
| 640 | 10 | 20 | 3 | 9 | N/A | 10 |
| 660 | 13 | 22 | 3 | 12 | N/A | 10 |
| 680 | 20 | 33 | 3 | 16 | N/A | 11 |
| 700 | 26 | 47 | 5 | 20 | N/A | 14 |
| 720 | 32 | 57 | 7 | 25 | N/A | 15 |
| 740 | 41 | 62 | 9 | 27 | N/A | 18 |
| 760 | 55 | 68 | 19 | 37 | N/A | 20 |
| 780 | 55 | 71 | 32 | 50 | N/A | 20 |
| 800 | 55 | 74 | 35 | 53 | N/A | 20 |
| 820 | 55 | 75 | 36 | 54 | N/A | 20 |
| 840 | 58 | 75 | 37 | 55 | N/A | 20 |
| 860 | 58 | 75 | 37 | 55 | N/A | 20 |

3.9 Physical requirements. The finished cloth shall conform to the requirements listed below, in Table VII, when tested as specified in 4.4.6.

TABLE VII. Physical requirements.

| Characteristic | All Classes |
|---|---|
| Weight, oz./sq.yd., (maximum) | 1.0 |
| Yarns per inch, (minimum)<br>WarpFilling | 120<br>120 |
| Breaking strength, pounds (minimum)<br>Warp<br>Filling | <br>42<br>42 |
| Tearing strength, pounds (minimum)<br>Warp<br>Filling | <br>5.0<br>5.0 |

3.9.1 Weave. The ripstop weave pattern for the cloth shall be one (1) of the weaves as shown in Figures 1, 2, or 3. Reinforcement ribs in both the warp and filling shall form a uniform pattern of squares. There shall be a minimum of 6.5 repeats of the pattern per inch in both directions.


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3.9.2 Width. For Government procurements only, the width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.10 Finish. The cloth shall be scoured and heat treated prior to printing to meet the requirement in 3.11.

3.11 Dimensional stability. The cloth shall show no appreciable distortion or puckering and the shrinkage or elongation shall not be greater than 2.0 percent in either warp or filling direction, when tested as specified in 4.4.6.

3.12 pH. The pH of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.6.

3.13 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.6. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used (see 6.7).

3.14 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 75 yards. Each length shall be put-up full width on a roll as specified in 5. 1.

3.15 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.16 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word “FACE” at each end of the roll.

3.17 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the applicable established maximum point values.

## 4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1.1, shall be inspected for conformance with physical requirements of Table VII, examined for appearance, color, defects in Table VIII, and overall workmanship. The presence of excessive defects, as defined in the contract (see 6.2) or failure of any testing shall be cause for rejection of the first article.


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4.3 Conformance inspection. In accordance with 4.1 above and 3.1.2, the material shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents. Conformance inspection shall include the examination of 4.4 and 4.4.6 (see 6.2).

4.4 Yard by Yard Examination. Each roll for Class 1, 2, 4, 5, 7, and 8 in the sample shall be examined yard-by-yard on the printed side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in ASTM 3990 that are clearly noticeable at normal inspection distance (3-feet) shall be scored and assigned demerit points as listed in 4.4.1 except that only those slubs and knots which exceed the limits shown on the Fabric Defect Replica Scale (see 2.3), "D" as applicable for slubs and "C" for knots, shall be scored and coarse yarn shall only be scored as a defect when the coarse yarn is twice the diameter of the normal yarn used in the fabric. No linear yard (increments of 1-yard on the measuring device of the inspection machine) from any one (1) roll shall be penalized more than four (4) points. The sample size shall be selected at random, no less than 10 percent of the rolls per lot. Acceptance criteria for visual examination shall be as specified in the contract or order (see 6.2).

Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.4.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3-inches in any dimension - one (1) point
For defects exceeding 3-inches, but not exceeding 6-inches in any dimension - two (2) points
For defects exceeding 6-inches, but not exceeding 9-inches in any dimension - three (3) points
For defects exceeding 9-inches in any dimension - four (4) points

4.4.2 Visual examination. The cloth shall be examined for the defects listed in Table VIII.

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TABLE VIII. Visual examination defects.

| Examine | Defect | Classification | |
|---|---|---|---|
| | | Major | Minor |
| Material defects and damages | Any hole, cut, tear smash, burn, exposed drill hole, thin place, needle chew, visible mends, broken or missing yarn. | 101 | 201 |
| | Knots greater than Fabric Defect Scale Level C (see 2.3). | 102 | |
| | Slubs greater than Fabric Defect Scale Level D (see 2.3). | 103 | |
| | Uneven weaving throughout. | 104 | |
| | Area of poor dye penetration, dye streak, or shade bar Colors not as specified (face and back). | 105 | |
| | Design not printed on face side of cloth. | 106 | |
| | Marking on face side of cloth omitted. | 107 | |
| | Width not as specified within a tolerance of ±1/2-inch. | | |
| | Camouflage pattern (face side) | | |
| | Any skitteriness of pattern exceeding that shown by the standard sample. | 108 | |
| | Pattern design not equal to standard sample. | 109 | |
| | Excessive feathering or spew of pattern. | 110 | |
| | Pattern repeat not equal to the standard sample. | 111 | |
| | Warpwise pattern repeat: | | |
| | Woodland Camouflage pattern (Class 4) less than 24.75- inches or more than 28.50-inches. | 112 | |
| | Desert Camouflage pattern (Class 5) less than 14.50- inches or more than 18.00-inches. | 113 | |
| | Operational Camouflage pattern (Class 7) less than 22.75- inches or more than 26.50-inches. | 114 | |
| | Woodland MARPAT (Class 8 less than 33.50 inches or more than 36.50 inches | 115 | |
| | Excessive grinning (off register, gap where ground shade shows through) of pattern as compared to the standard sample. | 116 | |
| | Any color off shade, not uniform, mottled, or spotted (face side only) | 117 | |
| Cleanliness | Any spot, streak, or stain of a permanent nature. | | 202 |
| | Removable spot, streak, or stain on outside of garment. | | 203 |
| | Any embedded foreign matter in the cloth.1/ | | 204 |
| | Objectionable odor. | | 205 |

1/ Clearly visible at normal inspection distance 3-feet.

4.4.3 and 4.4.4 DELETED

4.4.5 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined that each roll is labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act (see 3.15).



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4.4.6 End item testing. The cloth shall be tested for the characteristics listed in Table IX. The methods of testing as specified wherever applicable and as listed in Table IX shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more sample units or the lot average for dimensional stability fail to meet any requirement specified. The sample size shall be in accordance with the following:

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

4.4.6.1 Basic cloth. The basic cloth shall be tested in accordance with Table IX.

TABLE IX. End item tests.

| Characteristic | Requirement Reference | Test Method |
|---|---|---|
| Fiber identification | 3.5.1 | AATCC TM20 |
| Visual shade matching (all classes) | 3.6.1 | 1/ |
| Colorfastness:<br>Laundering (after 3 cycles)<br>Perspiration (acid & alkaline)<br>Crocking (wet & dry) | <br>3.6.2, Table I<br>3.6.2, Table I<br>3.6.2, Table I | <br>AATCC TM61 Test 1A and 4.5.2, 2/, 3/<br>AATCC TM15 and 4.5.2, 2/, 3/<br>AATCC TM8 and 4.5.2, 4/ |
| Dyeing/printing of cloth | 3.6.3 - 3.6.9 | Visual |
| Pattern execution | 3.7 | 4.5.3 |
| Pattern width | 3.7.1 | ASTM F2203 |
| Spectral reflectance:<br>Class 1 (Solids)<br>Class 4<br>Class 5<br>Class 7<br>Class 8 | <br>3.8.1, Table II-A<br>3.8.1.1, Table II-B<br>3.8.2, Table III<br>3.8.4, Table IV<br>3.8.5, Table VI | <br>4.5.4<br>4.5.4<br>4.5.4<br>4.5.4<br>4.5.4 |
| Weight | 3.9, Table VII | ASTM D3776/D3776M |
| Yarns per inch | 3.9, Table VII | ASTM D3775 |
| Breaking strength | 3.9, Table VII | ASTM D5035 |
| Tearing strength | 3.9, Table VII | ASTM D2262, 5/ |
| Weave | 3.9.1 | Visual |
| Dimensional stability | 3.11 | 4.5.1 |
| pH | 3.12 | AATCC TM81 |
| Toxicity | 3.13 | 4.5.5 |



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1/ The color and appearance of the dyed and printed cloth shall match the standard sample when viewed using the AATCC EP9, Option C see (6.7), with sources simulating artificial daylight $D_{75}$ illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).
2/ Only the stain on the nylon fibers of the color transfer cloth shall be evaluated.
3/ Rated using the AATCC EP1, Gray Scale for Color Change and AATCC EP2, Gray Scale for Staining.
4/ Rated using the AATCC EP8, AATCC 9-Step Chromatic Transference Scale.
5/ The average of five (5) highest peaks shall be used.

4.5 Methods of inspection. All testing shall be done in a standard condition environment defined by ASTM D1776/D1776M, if not specifically defined by the individual test procedure.

4.5.1 Dimensional stability test. The test specimen shall be a square of cloth a minimum 20- inches by 20-inches. Before marking and measuring for dimensional change, the specimen shall be in equilibrium with Standard Conditions as specified in ASTM D1776/D1776M. The specimen shall be laid without tension on a flat surface and shall be inscribed with an 18-inch square approximately equidistant from the edges having sides parallel to the warp and normal filling directions. The marked specimen shall be placed in a preheated oven and maintained at a temperature of 280 (±20 ºF) for a period of 2-hours. The specimen may be looped over a rod or laid flat on a rack both of which shall be made of low-conductivity (nonmetallic) material and have free air circulation on both sides. Promptly remove the specimen from the oven, allow to cool on a flat surface, and then bring to equilibrium under Standard Conditions. Measure each side of the inscribed square for dimensional change in both the warp and filling directions (two (2) measurements each for the warp and filling per specimen). Each change in dimension shall be reported to the nearest 0.1 percent. The specimen shall also be visually compared with the original unheated cloth for any appreciable distortion or puckering. (Appreciable means a change that is immediately noticeable when comparing the tested specimen with the original. If closer inspection is required to make apparent a slight change, the change is not considered appreciable).

4.5.2 Colorfastness evaluation testing. When testing for colorfastness properties, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicated which colors were evaluated together.

4.5.2.1 Operational Camouflage Pattern (OCP) (Class 7). All Colorfastness testing for Operational Camouflage Pattern (OCP), Class 7, shall be performed on the solid color area and not the tonal area.

4.5.3 Pattern execution. The pattern of the cloth shall be matched to the specified pattern drawing (see 2.2.2).


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4.5.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for Classes 1, 4, 7, and 8, and 700 to 860 nanometers (nm) for Class 5 at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g. magnesium oxide or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with layers of the same fabric and shade as follows: The cloth for Class 1, shall be measured as a single layer using three (3) backing layers of the same shade. The cloth for Class 4, shall be measured as a single layer using three (3) backing layers of the same shade for Light Green 354, Dark Green 355 and Brown 356 colors. The cloth for Class 5 shall be measured as a single layer backed with four (4) layers of the same shade cut from the standard. The cloth for Class 7, shall be measured as a single layer backed with eight (8) layers of the same shade. The cloth for Class 8 shall be measured as a single layer backed with six (6) layers of the same shade. The specimen shall be viewed at an angle no greater than 10 degrees from normal, with the specular component included. Measurements shall be taken on a minimum of two (2) different areas. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns when presented to the sample port. Photometric accuracy of the spectrophotometer shall be within 1-percent and wavelength accuracy within 2-nanometers. The diameter for standard aperture size used in the color measurement device shall be the largest possible for each color. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.5.5 Toxicity testing. Unless otherwise specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of these studies indicate the cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.13) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.


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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use as the outer covering for batting used in the manufacture of quilted clothing. The cloth is not intended for the manufacture or repair of parachutes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of cloth required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1.1, 4.2 and 6.3)
e. Conformance inspection acceptance quality limits (AQL) (see 3.1.2 and 4.3)
f. Camouflage pattern drawing, if required (see 3.5)
g. When toxicity testing is required (see 3.13 and 4.5.5)
h. Inspection conditions (see 4.4)
i. Packaging (see 5.1)

6.3 First article. When a first article inspection is required (see 3.1.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

6.6 Supersession data. The Class 3 U.S Army Camouflage Pattern 1948 and Class 6 Universal Camouflage Pattern cloth have been deleted since they are no longer required.

6.7 EPA information is available online at https://www.epa.gov.

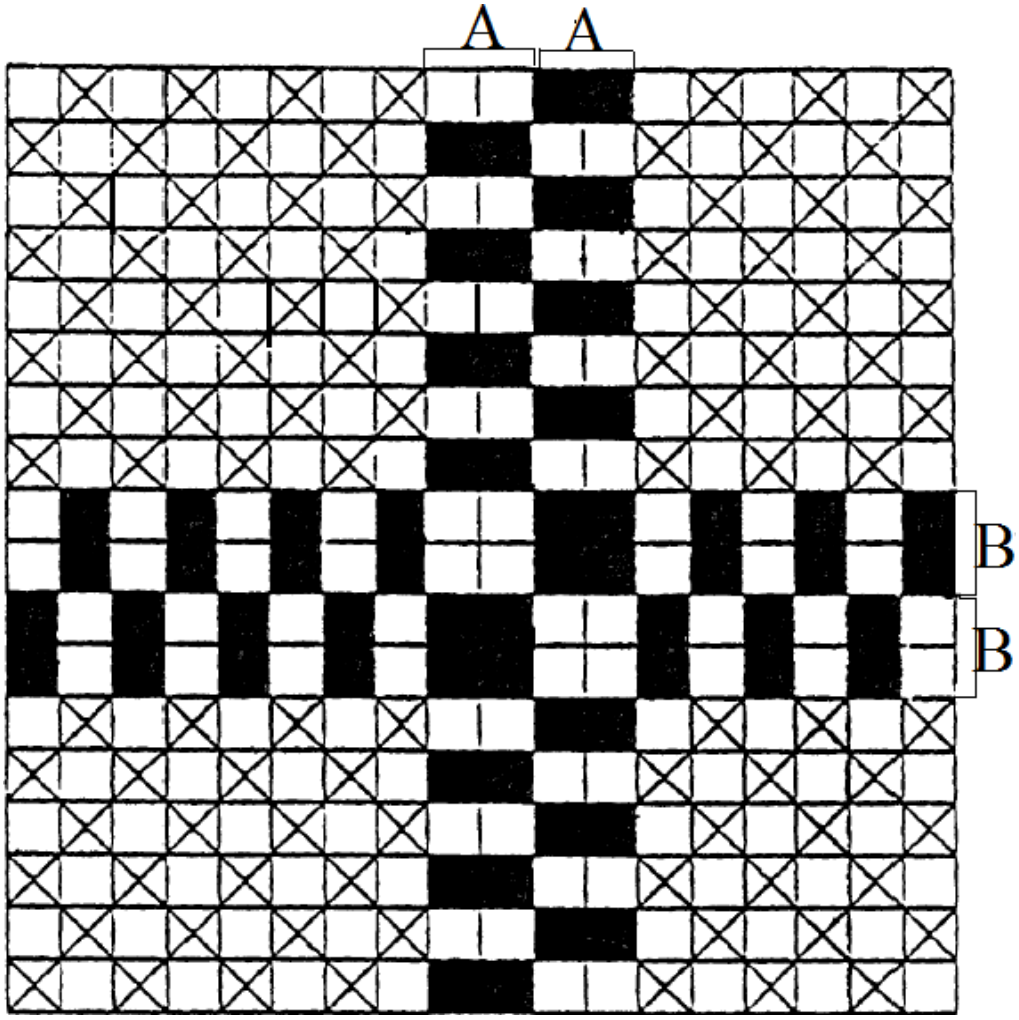
6.8 Subject term (key word) listing.

All Purpose Liner
Batting
Desert Camouflage
Pattern Liner
Operational Camouflage Pattern (OCP)



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Poncho
Quilted clothing
Woodland Camouflage Pattern
Woodland, MARPAT

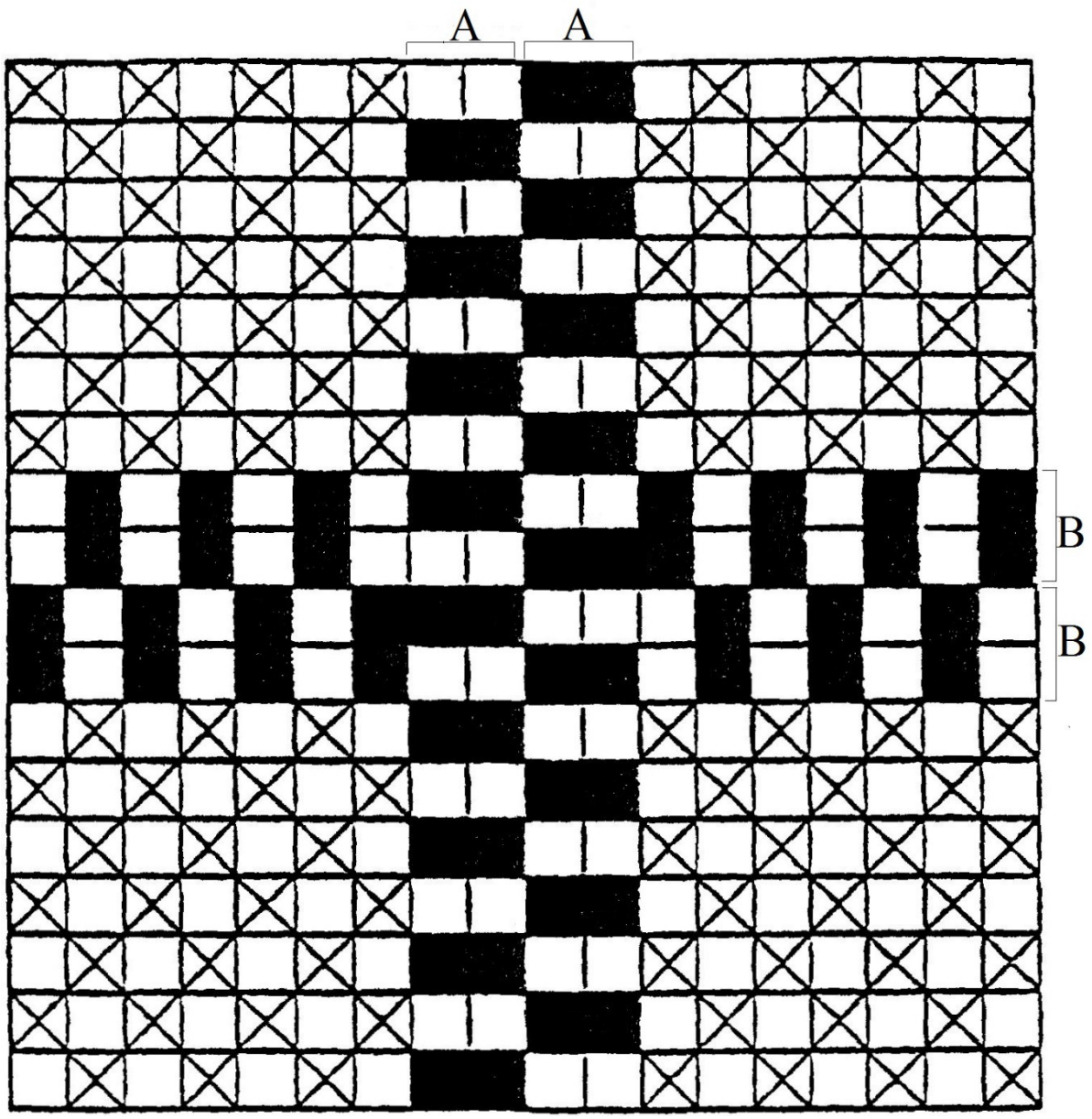



A = Two warp ends woven as one
B = Two filling picks per shed

FIGURE 1. Ripstop weave pattern.



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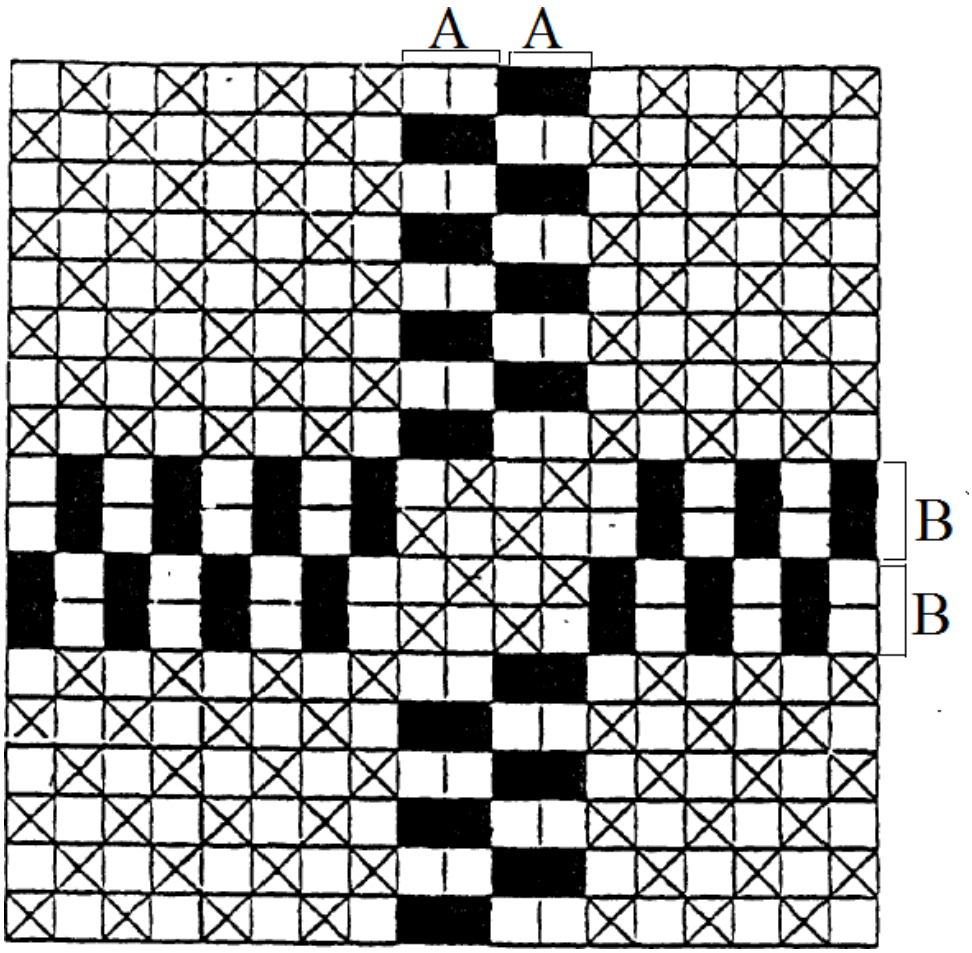

A =Two warp ends woven as one

B = Two filling picks per shed

FIGURE 2. Ripstop weave pattern.



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A = Two warp ends woven as one
B = Two filling picks per shed

FIGURE 3. Ripstop weave pattern.

Custodian:
Army – GL
Navy – NU
Air Force – 11

Preparing activity:
DLA - CT

Review activities:
Army MD
Navy – MC

(Project: 8305-2021-015)

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